$$\left[\begin{array}{ccc|ccc} 1 & \frac{-1}{2} & \frac{1}{4} & \frac{1}{4} & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 9 & 3 & 1 & 0 & 0 & 1 \end{array}\right] \begin{array}{l} A_2 = \\ B_2 = \\ C_2 = \end{array}$$